

THE HARBORTH CONSTANT OF $C_9 \oplus C_9$

ABSTRACT. For a finite abelian group G , let $\mathbf{g}(G)$ be the least integer k such that every subset of G with at least k elements contains an $\exp(G)$ -element zero-sum subset. We prove

$$\mathbf{g}(C_9 \oplus C_9) = 17.$$

An explicit 16-element set gives the lower bound. The upper bound is an exhaustive computer-assisted verification after affine normalization to sets containing $\{(0, 0), (1, 0)\}$. The computation uses exact subsum states, canonical prefixes under a stabilizer of order 108, and exact integer accounting of every searched or pruned completion. This confirms the $n = 9$ case of the Gao–Thangadurai conjecture.

1. INTRODUCTION

Let C_n denote the cyclic group of order n . Motivated by Harborth’s lattice-point problem [3], the Harborth constant of a finite abelian group G is

$$\mathbf{g}(G) = \min \left\{ k : \begin{array}{l} \text{every } A \subseteq G \text{ with } |A| \geq k \text{ contains } B \subseteq A, \\ |B| = \exp(G) \text{ and } \sum_{b \in B} b = 0 \end{array} \right\}.$$

Gao and Thangadurai conjectured [1] that

$$(1) \quad \mathbf{g}(C_n \oplus C_n) = \begin{cases} 2n - 1, & n \text{ odd,} \\ 2n + 1, & n \text{ even.} \end{cases}$$

The rank-two problem remains open in general; see [2, Section 1.1]. The parameter $n = 9$ is the first odd composite case, and to the best of our knowledge its exact value was not previously determined.

Theorem 1. *Every 17-element subset of $C_9 \oplus C_9$ contains a 9-element zero-sum subset, whereas some 16-element subset contains none. Consequently,*

$$\mathbf{g}(C_9 \oplus C_9) = 17.$$

The lower bound is elementary. The upper bound is the exact exhaustive verification described below, which is recomputable from the data printed in Section 3.

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2. LOWER BOUND AND AFFINE NORMALIZATION

Throughout, write

$$R = \mathbb{Z}/9\mathbb{Z}, \quad G = R^2, \quad e_1 = (1, 0).$$

Proposition 2. *One has $g(G) \geq 17$.*

Proof. Let

$$A = \{0, 1, \dots, 7\} \times \{0, 1\}.$$

The set A has 16 elements. Any 9-element subset of A contains t points with second coordinate 1, where $1 \leq t \leq 8$, because each horizontal layer has only eight points. Its second-coordinate sum is therefore nonzero modulo 9. Thus A contains no 9-element zero-sum subset. $\square$

Call $v \in G$ *primitive* if $v \notin 3G$.

Lemma 3. *Every subset of G with more than nine elements contains two points whose difference is primitive.*

Proof. The subgroup $3G$ has order 9. If every pairwise difference in a set S belonged to $3G$, then, after fixing $s_0 \in S$, one would have

$$S \subseteq s_0 + 3G,$$

and hence $|S| \leq 9$. $\square$

Lemma 4. *If a 17-element subset of G without a 9-element zero-sum subset exists, then one exists containing*

$$B = \{0, e_1\}.$$

Proof. By Lemma 3, such a set contains a, b for which $b - a$ is primitive. At least one coordinate of $b - a$ is a unit in R , so there is an $M \in \text{GL}_2(R)$ with

$$M(b - a) = e_1.$$

The affine map $x \mapsto M(x - a)$ sends a, b to $0, e_1$.

For any affine automorphism $\phi(x) = Mx + c$ and any nine points $x_1, \dots, x_9$,

$$\sum_{i=1}^9 \phi(x_i) = M \sum_{i=1}^9 x_i + 9c = M \sum_{i=1}^9 x_i.$$

Since M is invertible, a nine-term sum is zero if and only if its image sum is zero. $\square$

3. EXHAUSTIVE VERIFICATION

Set $E = G \setminus B$ and order its 79 elements increasingly by the integer representative $x + 9y$. Write

$$E = \{q_0, q_1, \dots, q_{78}\}$$

in this order. A normalized candidate is $B \cup T$ with

$$T \in \binom{E}{15}.$$

3.1. Exact subsum states. For a partial selected set $P \subseteq G$, define

$$\Sigma_j(P) = \left\{ \sum_{x \in U} x : U \subseteq P, |U| = j \right\}, \quad 0 \leq j \leq 8,$$

with $\Sigma_0(P) = \{0\}$.

Lemma 5. *Assume that P contains no 9-element zero-sum subset. A point $z \notin P$ may be added while preserving this property if and only if*

$$-z \notin \Sigma_8(P).$$

After adding z ,

$$\Sigma_j(P \cup \{z\}) = \Sigma_j(P) \cup (z + \Sigma_{j-1}(P)), \quad 1 \leq j \leq 8.$$

Proof. Any newly created 9-element zero-sum subset must contain z , and its other eight points must sum to $-z$. The update formula partitions the j -element subsets according to whether they contain z . $\square$

The verifier represents each $\Sigma_j(P)$ by an exact 81-bit set and updates the states in descending order of j .

3.2. Canonical prefixes. Let

$$\Gamma = \text{Stab}_{\text{AGL}_2(R)}(B)$$

be the setwise affine stabilizer of B . Its elements are exactly the maps

$$(x, y) \mapsto (x + ay, uy) \quad \text{and} \quad (x, y) \mapsto (1 - x + ay, uy),$$

where $a \in R$ and $u \in R^\times$. Hence

$$|\Gamma| = 2 \cdot 9 \cdot 6 = 108.$$

Let π_γ be the permutation of the positions $\{0, \dots, 78\}$ induced by $\gamma \in \Gamma$.

For $p = (p_1, p_2, p_3)$ with $p_1 < p_2 < p_3$, call p *canonical* if, for every $\gamma \in \Gamma$, the increasing rearrangement of

$$(\pi_\gamma(p_1), \pi_\gamma(p_2), \pi_\gamma(p_3))$$

is lexicographically at least p .

Lemma 6. *Every Γ -orbit in $\binom{E}{15}$ contains a set whose first three positions form a canonical triple.*

Proof. Choose the lexicographically least set

$$T = \{t_1 < \dots < t_{15}\}$$

in its orbit. If (t_1, t_2, t_3) were not canonical, some $\gamma \in \Gamma$ would send these three positions, after sorting, to a triple smaller than (t_1, t_2, t_3) . The first three positions of $\gamma(T)$ are lexicographically no larger than that sorted triple, contradicting the minimality of T . $\square$

Exact enumeration gives 992 canonical triples. A triple can be the first three positions of a 15-element set only if $p_3 \leq 66$, since twelve larger positions must remain. Exactly 831 canonical triples satisfy this condition; these are the *feasible canonical prefixes*. The exact number of completions below them is

$$(2) \quad \sum_{\substack{p \text{ canonical} \\ p_3 \leq 66}} \binom{78 - p_3}{12} = 811,509,569,804,714.$$

The prefixes need not be unique orbit representatives; Lemma 6 is used only to ensure that no orbit is omitted.

3.3. Search and verification. At a search node, let P be the current partial set and suppose that r further points are required. The verifier uses the following exact rejection rules.

- (1) A remaining point z is individually inadmissible exactly when

$$-z \in \Sigma_8(P),$$

by Lemma 5.

- (2) If fewer than r remaining points are individually admissible, no completion exists.
- (3) On the individually admissible points, form a graph in which distinct u, v are adjacent when

$$-(u + v) \notin \Sigma_7(P).$$

Every valid r -point completion must induce an r -clique: a nonedge u, v , together with seven points of P , would form a 9-element zero-sum subset. The verifier applies iterative degree peeling, which preserves every r -clique, followed by an exact clique search, and rejects a node only when no r -clique exists.

The depth-first search partitions the lexicographic completion family at each recursive node into disjoint subfamilies. When a subfamily is rejected, its exact cardinality is credited using an integer binomial coefficient. Thus every completion counted in (2) is either examined or belongs to a soundly rejected subtree. The final aggregate is as follows.

Quantity	Value
Affine stabilizer size	108
Canonical triples	992
Feasible canonical prefixes	831
Candidate completions	811,509,569,804,714
Depth-first-search nodes	879,672,298
Completions covered	811,509,569,804,714
Valid completion found	no

Proposition 7. *There is no $T \in \binom{E}{15}$ for which $B \cup T$ contains no 9-element zero-sum subset.*

Proof. The exact search rejected every completion below every feasible canonical prefix, and the covered count equals the independently computed total in (2). By Lemma 6, every Γ -orbit of normalized candidates has a searched representative. $\square$

Exact verification. Section 3 fixes the computation completely, so a referee can recompute every number above from this paper alone. Ordering E by $x + 9y$ regenerates the 108 maps of Γ and the position permutations π_γ , the 992 canonical triples, the 831 of them with $p_3 \leq 66$, and the total (2); the search is then driven by Lemma 5 and the three rejection rules, each rejected subfamily being credited by an exact binomial coefficient, so that the covered count is recomputed and compared with (2). All arithmetic is exact integer and bitset arithmetic, with no floating point and no randomness, and the subsum states are self-tested against direct enumeration before the search begins. The computation was re-run independently on a separate machine, by a program written from these definitions rather than from the original code, and the six quantities of the table that do not depend on the traversal order agree with what is reported here: the stabilizer size 108, the 992 canonical triples, the 831 feasible prefixes, the total (2), its equality with the completions covered, and the absence of a valid completion; only the node count is specific to an implementation. That program accompanies this paper, together with the transcript of a run of it made under a bounded time budget: in the transcript the counts listed above are re-derived in full, while the census over the 831 feasible prefixes is carried only as far as the budget allows, the uncovered remainder being reported explicitly rather than passed over; the census is completed by running the program with no time budget. Nothing above depends on either file. The re-run is a second computation, not a proof.

Proof of Theorem 1. Proposition 2 gives the lower bound. If a 17-element set without a 9-element zero-sum subset existed, Lemma 4 would produce one containing B , contradicting Proposition 7. Hence

$$g(C_9 \oplus C_9) \leq 17,$$

and equality follows. $\square$

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